

PROJECT SHOWCASE

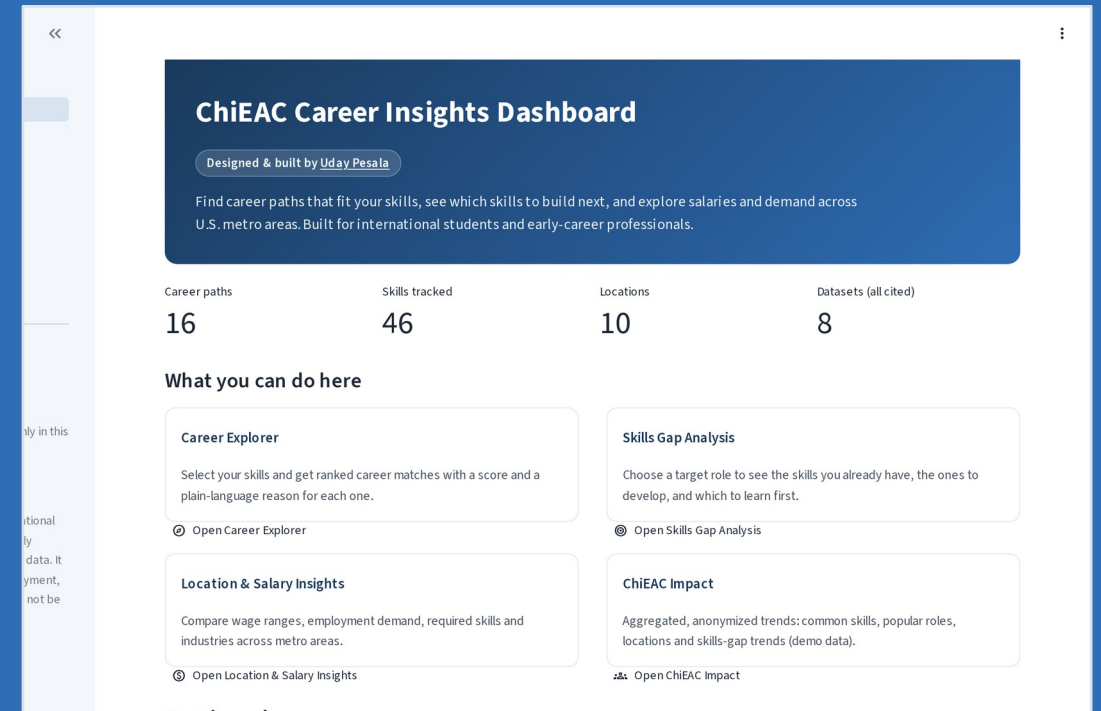
ChiEAC Career Insights Dashboard

Helping international students find careers that fit their skills, close skill gaps, and understand salaries and demand across U.S. cities.

Designed & built by Uday Pesala

Live app uday-pesala-chieac-project-career-insights-dashboard.streamlit.app

Built with Python · Streamlit · Pandas · Plotly · scikit-learn



The challenge, and what I built

THE CHALLENGE

- **Unclear job titles.** Strong skills, but no clear map to U.S. roles like Data Analyst or Cloud Engineer.
- **Scattered information.** Salary, demand and skills data live across many government sites.
- **Generic advice.** Students don't know which skills to learn first for their target role.

THE SOLUTION

- **Personal career matches** from your own skills, with a score and a plain-language reason.
- **A prioritized learning plan** showing what you have, what's missing, and what to learn first.
- **Pay and demand by city**, using cited public data (BLS, O*NET, Data USA), in one place.

16

career paths

46

skills tracked

10

U.S. metro areas

8

cited datasets

Four tools in one dashboard

Career Explorer

Pick skills, get ranked careers with a 0–100 match score and the reason why.

01

Skills Gap Analysis

Readiness score, skills to build by priority, and a downloadable PDF/CSV report.

02

Location & Salary Insights

Salary ranges, job demand and hiring industries across 10 metros.

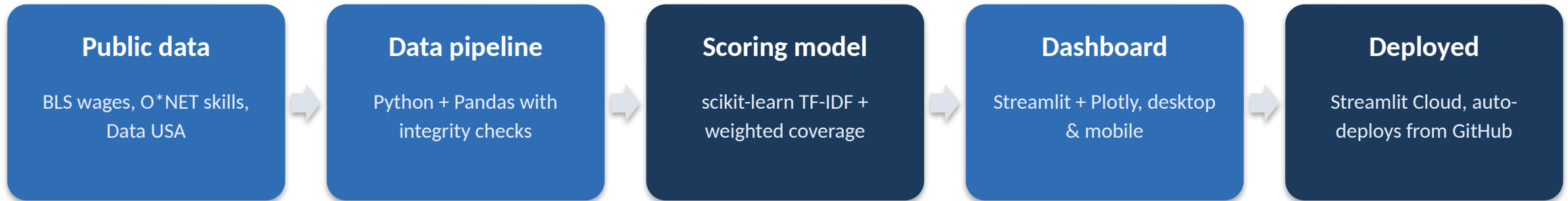
03

ChiEAC Impact

Anonymized community trends: top skills, target roles and skill gaps.

04

Built end to end: data, model, app and deployment



TRANSPARENT SCORING

Match = 60% skill coverage + 40% specificity

Coverage: how much of the role's weighted skill profile you have.

Specificity: TF-IDF similarity, so distinctive skills count more than common ones.

Priority = 70% importance + 30% transferability

High / Medium / Low, explained on every page.

BUILT TO BE TRUSTED

- **Cited data:** source & date under every chart; demo data clearly labelled.
- **Privacy first:** no accounts, nothing stored; small groups (<10) hidden.
- **Quality:** input validation, error handling, 17 automated tests.
- **Reproducible:** documented data dictionary and build script.

Value for ChiEAC, and what's next

For students

- Clear career options from their own skills
- A focused plan for what to learn next
- A report to bring to advisors

For advisors

- A shared, evidence-based starting point
- Explainable scores for 1:1 sessions
- Useful in workshops and events

For ChiEAC

- Anonymized insight into skills & gaps
- Data to plan workshops & partnerships
- A ready foundation for impact reporting

NEXT STEPS

- **Pilot** with a ChiEAC student cohort and gather feedback
- **Connect real program data** (with consent), replacing demo data
- **Refresh** with the latest BLS and O*NET releases

Try it live

uday-pesala-chieac-project-career-insights-dashboard.streamlit.app

github.com/Udaypesala123/Chicago-education-advocacy-cooperative

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